

Exercise Set II

Submission deadline: 22nd May 2026, 21:59

General notation

The following notation is used throughout this assignment:

S, S_t	underlying stock (asset) price; S_t denotes the price at time t .
K	strike price of the option.
r	risk-free interest rate (continuously compounded, annualised).
σ	volatility of the underlying asset's returns (annualised).
T	time to maturity of the option (in years).
t	discrete time step index (e.g. $t = 0, 1, 2, 3$ in the numerical example).
Δt	time-step size used in the discretisation.
X, Y	regression variables in the least-squares Monte Carlo procedure: X denotes the in-the-money stock price at the current exercise date, and Y denotes the corresponding discounted future cash flow.
$E[Y X]$	conditional expectation of Y given X , approximated by least-squares regression.
N	number of Monte Carlo simulation paths.
M	number of exercise opportunities (time steps) for the American option.
s.e.	standard error of the Monte Carlo estimate.
FD	Finite Difference method.
FTCS	Forward-Time Centred-Space scheme used in the provided <code>ftcs_bs_put.py</code> .

1. Least-squares Monte Carlo Simulation (12 points)

Study the original paper that introduces least-squares Monte Carlo simulation for pricing American options:

<https://people.math.ethz.ch/~hjfurrer/teaching/LongstaffSchwartzAmericanOptionsLeastSquareMonteCarlo.pdf>

- (i) Use the same parameter values provided in the “A Numerical Example” section of the paper, i.e. strike price is 1.1 and the risk-free interest rate is 6%. Use the same 8 paths from the paper as inputs to your Python script. Those 8 paths are also copied below in Figure 1.

Path	Stock price paths			
	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1.00	1.09	1.08	1.34
2	1.00	1.16	1.26	1.54
3	1.00	1.22	1.07	1.03
4	1.00	.93	.97	.92
5	1.00	1.11	1.56	1.52
6	1.00	.76	.77	.90
7	1.00	.92	.84	1.01
8	1.00	.88	1.22	1.34

Figure 1 Copy of the 8 paths used in the Numerical Example of the original least-squares Monte Carlo paper

Using Python, replicate the following results in the paper and include the replicated results in your answer sheet:

- Table “Cash-flow matrix at time 3”, Table “Regression at time 2”, the least-squares approximation formula for $E[Y|X]$ at time 2, and Table “Optimal early exercise decision at time 2”. Note: the paper displays values rounded to 2 or 4 decimal places. In your Python output, please display all numerical values with **5 digits** after the decimal point. **(2 points)**
 - Table “Cash-flow matrix at time 2”, Table “Regression at time 1”, the least-squares approximation formula for $E[Y|X]$ at time 1, and Table “Optimal early exercise decision at time 1”. In your Python output, please display all numerical values with **5 digits** after the decimal point. **(2 points)**
 - Table “Stopping rule” and Table “Option cash flow matrix”. In your Python output, please display all numerical values with **5 digits** after the decimal point. **(1 point)**
 - American put option price rounded to 5 decimal places. **(0.5 point)**
 - European put option price, using the same 8 paths, rounded to 5 decimal places. **(0.5 point)**
- (ii) Read the section “Valuing American Put Options” carefully, especially how Table 1 in that paper was generated.
- Use your script developed in (i) as the base. Replace the hard-coded paths by samples generated by Monte Carlo simulation, to prepare for the replication of Table 1. **(1.5 point)**
 - Replicate the columns included in the red box of Figure 2, and display **4 digits after the decimal point for all numbers**. Note: make sure that you are using the same

model settings and calculation settings as in the paper. (“Simulated American” **2.5 points**; “Closed form European” and “Early exercise Value” **1 point**; “s.e.” **1 point**)

Table 1

S	σ	T	Finite difference American	Closed form European	Early exercise Value	Simulated American	(s.e.)	Closed form European	Early exercise value	Difference in early exercise value
36	.20	1	4.478	3.844	.634	4.472	(.010)	3.844	.628	.006
36	.20	2	4.840	3.763	1.077	4.821	(.012)	3.763	1.058	.019
36	.40	1	7.101	6.711	.390	7.091	(.020)	6.711	.380	.010
36	.40	2	8.508	7.700	.808	8.488	(.024)	7.700	.788	.020
38	.20	1	3.250	2.852	.398	3.244	(.009)	2.852	.392	.006
38	.20	2	3.745	2.991	.754	3.735	(.011)	2.991	.744	.010
38	.40	1	6.148	5.834	.314	6.139	(.019)	5.834	.305	.009
38	.40	2	7.670	6.979	.691	7.669	(.022)	6.979	.690	.001
40	.20	1	2.314	2.066	.248	2.313	(.009)	2.066	.247	.001
40	.20	2	2.885	2.356	.529	2.879	(.010)	2.356	.523	.006
40	.40	1	5.312	5.060	.252	5.308	(.018)	5.060	.248	.004
40	.40	2	6.920	6.326	.594	6.921	(.022)	6.326	.595	-.001
42	.20	1	1.617	1.465	.152	1.617	(.007)	1.465	.152	.000
42	.20	2	2.212	1.841	.371	2.206	(.010)	1.841	.365	.006
42	.40	1	4.582	4.379	.203	4.588	(.017)	4.379	.209	-.006
42	.40	2	6.248	5.736	.512	6.243	(.021)	5.736	.507	.005
44	.20	1	1.110	1.017	.093	1.118	(.007)	1.017	.101	-.008
44	.20	2	1.690	1.429	.261	1.675	(.009)	1.429	.246	.015
44	.40	1	3.948	3.783	.165	3.957	(.017)	3.783	.174	-.009
44	.40	2	5.647	5.202	.445	5.622	(.021)	5.202	.420	.025

Comparison of the finite difference and simulation values for the early exercise option in an American-style put option on a share of stock, where the option is exercisable 50 times per year. The early exercise value is the difference between the American and European put values. In this comparison, the strike price of the put is 40, the short-term interest rate is .06, and the underlying stock price S , the volatility of returns σ , and the number of years until the final expiration of the option T are as indicated. The European option values are based on the closed-form Black-Scholes formula. The simulation is based on 100,000 (50,000 plus 50,000 antithetic) paths for the stock-price process. The standard errors of the simulation estimates (s.e.) are given in parentheses.

Figure 2 Copy of Table 1 in the original paper; the columns to be reproduced by you are highlighted in the red box.

2. Finite Difference Method (8 points)

Use the Python code “`ftcs_bs_put.py`”, which was provided to you before Lecture 10 (Finite Difference (FD) methods for option pricing), as the starting point.

- Adjust the code, such that it can produce European put prices for the same testing set as included in “Table 1” of the Least-squares MC paper (see also Figure 2 above). Please use the same discretization settings for the FD method as used in that paper. Print out the FD European option prices and compare them to your column “Closed form European” you produced in Question 1-(ii). (**2 points**)
- Modify the code in (i) to yield another Python script that prices American put options using Finite Difference method(s). In your answer sheet, include a snapshot of the lines

where the changes are made (**2 points**). Design some tests to demonstrate that your implementation is correct. (**2 points**)

- (iii) Let your Python code output the columns which are **not** included in the red box of Figure 2. In other words, with this step, you'll fully replicate the whole Table 1 of the paper. (**2 points**)